

CBSE Practice Worksheet

Class 7 | Mathematics | Difficulty: Hard | Set C

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the cube root of -125?

- (A) 5
- (B) -5
- (C) ± 5
- (D) undefined

2. If $2z = 3x = 6y$, which is true?

- (A) $1/z = 1/x + 1/y$
- (B) $z = x + y$
- (C) $z = xy$
- (D) $z = x - y$

3. Find: $(-1)^1 + (-1)^1$

- (A) 0
- (B) 2
- (C) -2
- (D) 1

4. Solve: $3(x-2)/4 = (2x+1)/3$

- (A) $x = 14$
- (B) $x = 22$
- (C) $x = 26$
- (D) $x = 30$

5. Solve: $\sqrt{x+7} = x-5$

- (A) $x = 9$
- (B) $x = 2$
- (C) $x = 9$ or 2
- (D) No solution

6. Find: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$

- (A) 2
- (B) 8
- (C) 15
- (D) $\sqrt{15}$

7. Factorize: $x^2 - 5x - 14$

- (A) $(x-7)(x+2)$
- (B) $(x+7)(x-2)$
- (C) $(x-7)(x-2)$
- (D) $(x+7)(x+2)$

8. Simplify: $(x^3 - 8)/(x - 2)$

- (A) $x^2 + 2x + 4$
- (B) $x^2 - 2x + 4$
- (C) $x^2 + 4$

(D) $x^2 - 4$

9. Solve: $|2x - 5| = 9$

(A) $x = 7$ or -2

(B) $x = 7$ or 2

(C) $x = -7$ or 2

(D) $x = -7$ or -2

10. Simplify: $(a^2 - b^2)/(a + b)$

(A) $a - b$

(B) $a + b$

(C) $a^2 - b$

(D) $a - b^2$

Section B: Fill in the Blanks (1 mark each)

11. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 = \underline{\hspace{2cm}}$

12. $\sqrt[3]{-64} = \underline{\hspace{2cm}}$

13. If $\log x = 2$, then $x = \underline{\hspace{2cm}}$

14. If $x^2 - 6x + 8 = 0$, then $x = \underline{\hspace{2cm}}$ or $\underline{\hspace{2cm}}$

15. $\text{HCF} \times \text{LCM}$ of 12 and 18 = $\underline{\hspace{2cm}}$

16. If $5^x = 125$, then $x = \underline{\hspace{2cm}}$

17. $(x + 1/x)^2 - (x - 1/x)^2 = \underline{\hspace{2cm}}$

18. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = \underline{\hspace{2cm}}$

Section C: Short Answer Questions (2 marks each)

19. Solve: $|x - 3| + |x + 2| = 7$

20. Solve: $x^2 - 7x + 12 = 0$

21. Factorize: $8x^3 - 27$

22. Simplify: $(x^2 - 16)/(x^2 + 4)$

23. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

24. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

25. If $x = 2 + \sqrt{3}$, find $x + 1/x$

--- End of Worksheet ---

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